
Artificial Intelligence

BS (CS) Spring 2026

Unsupervised Learning



Learning Objectives:

1. K-Mean Clustering

Lab Tasks

Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB14* e.g. **i23XXXX_A_LAB14**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Question

Question 1: K-Mean Clustering and Interpretation

Pumpkin Seeds Dataset

Using the Pumpkin Seeds Dataset, you are required to perform a complete K-Means Clustering analysis by completing all parts of this exam in a single coherent notebook. Each part builds on the previous one.

- (a) Load the dataset and extract all numerical features. Apply `StandardScaler` to normalize the data. Display the first 5 rows of both the original and scaled dataframes, and plot side-by-side box plots to justify why scaling is necessary before applying K-Means clustering.
- (b) Using the scaled data from part (a), apply the **Elbow Method** by computing WCSS for $k = 1$ to 10. Plot the elbow curve and state the optimal value of k with justification.
- (c) Train a K-Means model using the optimal k identified in part (b) on the scaled dataset. Map the resulting cluster labels to the actual class names (Çerçvelik and Ürgüp Sivrisi) and evaluate the model by reporting the **Accuracy**, **Confusion Matrix**, and **Adjusted Rand Index (ARI)**.
- (d) Compute the **Silhouette Score** for $k = 2$ to 6 using the scaled data. Plot the scores as a bar chart and compare the best k from this analysis against the result you obtained in part (b). Do both methods agree?
- (e) Apply **PCA** to reduce the dataset to 2 principal components. Train a K-Means model ($k=2$) on the PCA-reduced data and create a scatter plot with two subplots side by side — one showing K-Means cluster assignments and the other showing the actual class labels. Comment on how well the clusters align with the true classes.

(f) Based on all results from parts (a) through (e), write a **brief conclusion (5–7 sentences)** as a markdown cell in your notebook discussing: whether K-Means was an effective algorithm for this dataset, what the evaluation metrics revealed, and one limitation of K-Means that was evident from your analysis.